

Chapter 8: Supervised Learning - Model Performance Evaluation

Comprehensive Tutorial Notes: Continuous Metrics, Classification Metrics, ROC/AUC, Threshold Selection

GARP RAI Exam Preparation

T1: Model Evaluation Purpose

TOPIC: Primary Purpose of Model Performance Evaluation

DEFINITION:

- Model performance evaluation is the systematic process of assessing how well a supervised learning model generalizes to new, unseen data
- It provides the technical basis to understand model accuracy and the right questions to ask when choosing appropriate performance metrics

KEY CONCEPTS:

- 1 **Generalization Assessment:** How well does the model perform on data it has not seen during training?
- 2 **Model Comparison:** Which model is best for the specific problem at hand?
- 3 **Error Quantification:** How wrong can we afford to be, and what are the consequences when models are wrong?
- 4 **Risk Management:** What are the financial and reputational implications of model errors?

WHY IT MATTERS:

- Models drive critical decisions in credit risk, fraud detection, AML compliance
- Poor models have significant financial and reputational consequences
- Regulatory requirements demand rigorous model validation
- Stakeholder confidence depends on reliable model performance

PRACTICAL EXAMPLE:

- Greek bank estimated recovery rate at 65%
- Actual recovery was 40%
- Result: €120 million capital add-on, stock price slide, regulatory scrutiny

EXAM TRAP: Model evaluation is about **generalization performance**, not training performance!

T2: Continuous vs Classification Outputs

TOPIC: Distinguishing Between Continuous and Discrete Output Metrics

DEFINITION:

- Supervised learning models produce either continuous numerical outputs (regression) or discrete categorical outputs (classification)
- Each type requires fundamentally different evaluation approaches and metrics

CONTINUOUS OUTPUTS (REGRESSION):

- **Predict:** Numbers such as VaR, market prices, net revenue, expected credit losses
- **Metrics:** MSE, RMSE, MAE, MAPE
- **Key Question:** How numerically close are predictions to actual values?

DISCRETE OUTPUTS (CLASSIFICATION):

- **Predict:** Categories such as fraud vs legitimate, default vs paid off, approve vs decline
- **Metrics:** Accuracy, Precision, Recall, F1 Score, AUC
- **Key Question:** How well does the model distinguish between classes?

WHY THE DISTINCTION MATTERS:

- Continuous metrics measure numerical accuracy (how far off is the prediction?)
- Classification metrics measure classification accuracy (how many correct classifications?)
- Different types of errors have different costs in each context

EXAM TRAP: Continuous vs discrete outputs require fundamentally different evaluation approaches!

T3: Consequences of Poor Modeling

TOPIC: Real-World Consequences of Poor Model Performance

THE CHAIN OF CONSEQUENCES:

1 Financial Impact:

- Capital add-ons (e.g., €120 million for Greek bank)
- Shareholder dilution
- Underpricing of risk in new loan originations
- Future losses in portfolio

2 Operational Impact:

- Deep-dive reviews of entire credit risk modeling suite
- Significant internal resource consumption
- Potential systemic weaknesses requiring costly overhauls

3 Reputational Impact:

- Lingering drag on investor confidence
- Damaged relationships with regulators
- Market confidence erosion

4 Regulatory Impact:

- ECB supervisory review actions
- Increased scrutiny of all models
- Potential fines and sanctions

EXAM TRAP: Poor modeling has **real** financial, operational, and reputational consequences!

T4: Test Sample Importance

TOPIC: Why Compute Metrics on Test Sample

DEFINITION:

- The test sample comprises observations not used in determining parameter estimates or tuning hyperparameters
- It provides an unbiased estimate of how the model will perform on new, unseen data

THE OVERFITTING PROBLEM:

- Models learn training data patterns, including noise and random fluctuations
- Performance on training data is optimistically biased
- Training performance is not a true test of generalization
- A model that memorizes training data may fail completely on new data

DATA SPLIT PURPOSES:

- **Training Sample:** Estimate model parameters
- **Validation Sample:** Tune hyperparameters and select model architecture
- **Test Sample:** Evaluate final model performance (unseen data)

KEY INSIGHT:

- We are generally more interested in validation and test performance
- Examining accuracy on training data is not a true test of performance
- Test sample performance is what matters in practice

EXAM TRAP: Always evaluate on **test sample** for true generalization performance!

T5: Training vs Test Gap

TOPIC: Interpreting Training-Test Performance Gap

DEFINITION:

- The difference between model performance on training data versus test data
- A key diagnostic for detecting overfitting

PERFORMANCE PATTERNS:

- **No Gap:** Training performance $\approx$ Test performance (good generalization)
- **Small Gap:** Slight overfitting (acceptable in practice)
- **Large Gap:** Significant overfitting (problematic, needs remediation)

WHY THE GAP OCCURS:

- Model memorizes training patterns rather than learning generalizable relationships
- Learns noise instead of signal
- Too complex for the amount of training data available
- Insufficient regularization

EXAMPLE (LendingClub):

- Logistic Regression: Training accuracy = 84.2%, Test accuracy = 71.3%
- Neural Network: Training accuracy = 84.2%, Test accuracy = 70.1%
- Both show significant gap indicating overfitting

REMEDICATION:

- Remove irrelevant features
- Apply regularization (L1, L2, dropout)
- Get more training data
- Use cross-validation

EXAM TRAP: Large train-test gap = **overfitting!**

T6: Mean Squared Error (MSE)

TOPIC: Mean Squared Error - Definition and Calculation

FORMULA:

$$MSE = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

where:

- y_i = true value of outcome for observation i
- $\hat{y}_i$ = model prediction for observation i
- N = number of observations in test sample

WHAT IT MEASURES:

- Average of all squared forecast errors
- Squaring removes sign (under- and over-estimation penalized symmetrically)
- **Heavily penalizes large errors** (tail risk focus)

KEY CHARACTERISTICS:

- Always non-negative
- Lower is better (most accurate model has lowest MSE)
- Units are squared (e.g., dollars²) - difficult to interpret intuitively
- Sensitive to outliers

EXAMPLE (Salary Prediction):

- Model: $\hat{y} = 16.88 + 1.06 \times \text{experience}$
- Sum of squared errors = 431.02
- $N = 10$
- $MSE = 431.02 / 10 = 43.1$

EXAM TRAP: MSE uses **squared errors** - heavily penalizes large deviations!

T7: MSE Managerial Question

TOPIC: What Managerial Question Does MSE Answer?

MANAGERIAL QUESTION:

- "How do we penalize big errors in our model?"

MANAGERIAL FOCUS:

- Highlights tail risk
- Emphasis on squared errors heavily penalizes large deviations
- Sensitive to costly outliers
- Important for risk management

WHY THIS MATTERS:

- Large errors have significant consequences in risk management
- Risk managers care about tail events (extreme losses)
- Regulatory focus on extreme outcomes and capital adequacy
- MSE aligns with regulatory capital frameworks

REGULATORY PERSPECTIVE:

- ECB's Guide to Internal Models recommends metrics capturing tail risk
- Ensures large, infrequent losses are not overlooked
- RMSE reported alongside MAE for complete picture

EXAM TRAP: MSE is for **penalizing big errors** - risk managers love it!

T8: Root Mean Square Error (RMSE)

TOPIC: Root Mean Square Error - Definition and Interpretation

FORMULA:

$$RMSE = \sqrt{MSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2}$$

KEY CHARACTERISTICS:

- Same units as target variable (e.g., dollars)
- More interpretable than MSE
- Gives sense of average error magnitude
- Still sensitive to outliers due to squaring before square root

COMPARISON TO MSE:

- MSE: Units are squared (e.g., dollars²) - hard to interpret
- RMSE: Units are original (e.g., dollars) - easy to interpret
- Both penalize large errors disproportionately

EXAMPLE (Salary Prediction):

- $MSE = 43.1$
- $RMSE = \sqrt{43.1} = 6.57$
- Interpretation: "On average, predictions are off by about 6.57 thousand dollars"

CAVEAT FOR MANAGERS:

- Unusually large loss on single facility could disproportionately inflate RMSE
- Need to ask: "Is high RMSE driven by systemic issue or one or two extreme events?"

EXAM TRAP: RMSE is **easier to interpret** than MSE because it's in same units!

T9: RMSE Managerial Question

TOPIC: What Managerial Question Does RMSE Answer?

MANAGERIAL QUESTION:

- "What does the typical error feel like in our primary unit of currency (e.g., dollars)?"

MANAGERIAL FOCUS:

- Same units as target variable (e.g., dollars)
- Directly interpretable for non-technical stakeholders
- "On average, our predictions are off by RMSE dollars"
- Helps CEOs and board members understand model performance

WHY THIS MATTERS:

- Connects statistical measure to business impact
- Board members can relate to dollar figures
- Enables comparison across different models
- Facilitates communication with non-technical audiences

CAVEAT:

- Still sensitive to outliers (squaring before square root)
- An unusually large loss on single facility could disproportionately inflate RMSE
- May mask generally good performance across rest of portfolio
- Need to investigate if high RMSE is systemic or from extreme, explainable events

EXAM TRAP: RMSE answers "what does typical error feel like?"

T10: Mean Absolute Error (MAE)

TOPIC: Mean Absolute Error - Definition and Advantages

FORMULA:

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|$$

KEY CHARACTERISTICS:

- Uses absolute differences (not squared)
- Robust to outliers
- Easy to communicate
- Straightforward average error magnitude

LIMITATIONS:

- Can mask direction of bias (over vs under prediction)
- Doesn't heavily penalize occasional large misses
- May underestimate critical risk events from risk capital perspective

EXAMPLE (Salary Prediction):

- Sum of absolute errors = 50.11
- $N = 10$
- $MAE = 50.11 / 10 = 5.01$
- Interpretation: "On average, predictions are off by about 5.01 thousand dollars"

WHEN TO USE:

- When outliers are not critical
- When you want stable, interpretable measure
- When equal treatment of all errors is desired

EXAM TRAP: MAE is **robust to outliers** - unlike MSE and RMSE!

T11: MAE Managerial Question

TOPIC: What Managerial Question Does MAE Answer?

MANAGERIAL QUESTION:

- "On average, by how much are our predictions off, regardless of direction?"

MANAGERIAL FOCUS:

- "On average, our predictions are off by MAE dollars"
- Ignores direction (over vs under prediction)
- Easy to communicate to non-technical audiences

WHEN TO USE:

- When outliers are not critical
- When you want stable, interpretable measure
- When equal treatment of all errors is desired

COMPARISON WITH RMSE:

- $MAE = 5.01$ vs $RMSE = 6.57$ in salary example
- $RMSE \geq MAE$ because squaring penalizes larger errors more
- The gap between RMSE and MAE indicates presence of large errors

REGULATORY PERSPECTIVE:

- ECB recommends reporting RMSE alongside MAE
- MAE alone might mask critical risks
- Both together provide complete picture

EXAM TRAP: MAE answers "by how much are we off, regardless of direction?"

T12: Mean Absolute Percentage Error (MAPE)

TOPIC: Mean Absolute Percentage Error - Definition and Advantages

FORMULA:

$$MAPE = \frac{100}{N} \sum_{i=1}^N \left| \frac{y_i - \hat{y}_i}{y_i} \right|$$

KEY CHARACTERISTICS:

- Expressed as percentage
- Normalized - allows comparison across different scales
- Board-friendly due to percentage nature
- Consistent across portfolios

LIMITATIONS:

- Explodes or becomes undefined when actual values are zero or near zero
- Can be misleading if few small-value items have very high percentage errors
- May understate errors on large absolute value items

EXAMPLE (Salary Prediction):

- Sum of $|(y_i - \hat{y}_i)/y_i| = 1.62$
- $N = 10$
- $MAPE = 1.62 / 10 \times 100 = 16.2\%$
- Interpretation: "On average, forecast error is 16% of actual salary"

MOST INTUITIVE:

- Of all forecast accuracy measures, MAPE is most intuitive
- 16% can be interpreted as average forecast error is 16% of actual salary

EXAM TRAP: MAPE is **percentage-based** - great for communication but has limitations!

TOPIC: What Managerial Question Does MAPE Answer?

MANAGERIAL QUESTION:

- "In percentage terms, what is the typical error relative to the actual value?"

MANAGERIAL FOCUS:

- "On average, our predictions are off by MAPE%"
- Normalized - allows comparison across models/portfolios
- Perfect for audit committees and board presentations
- Common choice for auditors

WHEN TO USE:

- Comparing models with different scales
- Communicating to non-technical stakeholders
- When percentage errors are more meaningful than absolute

AUDITOR PREFERENCES:

- Auditors often gravitate to MAPE for pure percentage narrative
- Standardized measure for discussing model accuracy with audit committees
- Allows comparison across different models and portfolios

EXAM TRAP: MAPE answers "what's the error as a percentage?"

T14: MSE Calculation Example

TOPIC: Step-by-Step MSE Calculation

EXAMPLE DATA (Test Sample of 10 Observations):

i	x_i	y_i	$\hat{y}_i$	$y_i - \hat{y}_i$	$(y_i - \hat{y}_i)^2$
1	4	20.15	21.12	-0.97	0.94
2	6	21.48	23.24	-1.76	3.10
3	9	31.88	26.42	5.46	29.81
4	10	32.88	27.48	5.40	29.16
5	21	49.92	39.14	10.78	116.21
6	25	57.46	43.38	14.08	198.25
7	3	21.13	20.06	1.07	1.14
8	1	12.18	17.94	-5.76	33.18
9	2	19.47	19.00	0.47	0.22
10	7	28.66	24.30	4.36	19.01
Total					431.02

CALCULATION:

- Sum of squared errors = 431.02
- $N = 10$
- $MSE = 431.02 / 10 = 43.1$
- $RMSE = \sqrt{43.1} = 6.57$

EXAM TRAP: MSE = sum of squared errors divided by N!

T15: RMSE Interpretation

TOPIC: Interpreting RMSE in Context

EXAMPLE:

- Salary prediction model
- $RMSE = 6.57$
- Target variable measured in thousands of dollars

INTERPRETATION:

- "On average, predictions are off by about 6.57 thousand dollars"
- RMSE is in same units as target (salary in \$000s)
- Gives sense of typical error magnitude

IMPORTANT NUANCES:

- This is an average - some errors are larger, some smaller
- Squaring means larger errors contribute more to the average
- $RMSE = 6.57$ means typical error is around 6.57 units
- Not every prediction is off by exactly 6.57

COMPARISON WITH MAE:

- $MAE = 5.01$
- $RMSE = 6.57$
- Gap between RMSE and MAE indicates presence of larger errors
- If $RMSE \approx MAE$, errors are relatively uniform

EXAM TRAP: RMSE is in **same units as target** - easier to understand!

T16: MAE Calculation Example

TOPIC: Step-by-Step MAE Calculation

EXAMPLE DATA (Test Sample of 10 Observations):

i	x_i	y_i	$\hat{y}_i$	$y_i - \hat{y}_i$	$ y_i - \hat{y}_i $
1	4	20.15	21.12	-0.97	0.97
2	6	21.48	23.24	-1.76	1.76
3	9	31.88	26.42	5.46	5.46
4	10	32.88	27.48	5.40	5.40
5	21	49.92	39.14	10.78	10.78
6	25	57.46	43.38	14.08	14.08
7	3	21.13	20.06	1.07	1.07
8	1	12.18	17.94	-5.76	5.76
9	2	19.47	19.00	0.47	0.47
10	7	28.66	24.30	4.36	4.36
Total					50.11

CALCULATION:

- Sum of absolute errors = 50.11
- $N = 10$
- $MAE = 50.11 / 10 = 5.01$

INTERPRETATION:

- "On average, predictions are off by about 5.01 thousand dollars"
- Similar to RMSE but less influenced by large errors
- More stable measure of typical error

EXAM TRAP: MAE = average of absolute errors - sum divided by N!

T17: MAPE Calculation Example

TOPIC: Step-by-Step MAPE Calculation

EXAMPLE DATA (Test Sample of 10 Observations):

i	y_i	$\hat{y}_i$	$y_i - \hat{y}_i$	$\frac{y_i - \hat{y}_i}{y_i}$	$\frac{ y_i - \hat{y}_i }{y_i}$
1	20.15	21.12	-0.97	-0.05	0.05
2	21.48	23.24	-1.76	-0.08	0.08
3	31.88	26.42	5.46	0.17	0.17
4	32.88	27.48	5.40	0.16	0.16
5	49.92	39.14	10.78	0.22	0.22
6	57.46	43.38	14.08	0.25	0.25
7	21.13	20.06	1.07	0.05	0.05
8	12.18	17.94	-5.76	-0.47	0.47
9	19.47	19.00	0.47	0.02	0.02
10	28.66	24.30	4.36	0.15	0.15
Total					1.62

CALCULATION:

- Sum of $|(y_i - \hat{y}_i)/y_i| = 1.62$
- $N = 10$
- $\text{MAPE} = 1.62 / 10 \times 100 = 16.2\%$

INTERPRETATION:

- "On average, forecast error is 16% of actual salary"
- Most intuitive of all measures
- Easy to communicate to non-technical stakeholders

EXAM TRAP: MAPE = percentage of actual values - most intuitive!

TOPIC: Choosing Metrics to Capture Tail Risk

WHICH METRIC CAPTURES TAIL RISK?

- **MSE or RMSE** - because they square errors, heavily penalizing large deviations

WHY MSE/RMSE CAPTURE TAIL RISK:

- Squaring errors heavily penalizes large deviations
- Large errors contribute disproportionately to the metric
- Sensitive to outliers (tail events)
- Regulatory preference (ECB recommends)

COMPARISON:

- **MAE**: Robust, but ignores tail risk
- **MAPE**: Good for communication, but has limitations
- **MSE/RMSE**: Best for capturing extreme errors

REGULATORY PERSPECTIVE:

- ECB's Guide to Internal Models recommends metrics capturing tail risk
- Nudges firms to report RMSE alongside MAE
- Ensures large, infrequent losses are not overlooked

EXAM TRAP: ECB recommends metrics that **capture tail risk** - MSE and RMSE!

TOPIC: Appropriate Use Cases for MAPE

WHEN MAPE IS APPROPRIATE:

- Comparing models across different portfolios
- Comparing error on €100k loan vs €10m loan
- Communicating to audit committees
- When percentage errors are more meaningful than absolute

WHEN MAPE IS NOT APPROPRIATE:

- When actual values are near zero (MAPE explodes)
- When tail risk is the primary concern
- When absolute errors matter more than percentage

KEY ADVANTAGE:

- Normalized measure allows comparison across scales
- A €2,000 error on €10,000 loan (20%) vs €2,000 error on €1,000,000 loan (0.2%)
- MAPE treats these appropriately based on relative size

EXAM TRAP: MAPE is best for [comparing across different scales](#)!

TOPIC: Beyond Averages - The Story in Error Distributions

WHY LOOK BEYOND AVERAGES?

- Distribution of errors reveals patterns averages hide
- Senior leaders should encourage visualizations (histograms, box plots)
- Provides richer context for interpreting average performance metrics

KEY QUESTIONS TO ASK:

- 1 "Are our model errors generally small with a few large ones, or consistently off by moderate amount?"
- 2 "Are errors randomly distributed or is model systematically over-predicting (more negative errors) or under-predicting (more positive errors)?"
- 3 "Are there particular segments or conditions where errors are consistently larger or more biased?"

WHY THIS MATTERS:

- **Consistent bias** $\hat{\epsilon}$ random errors (more problematic even if average small)
- Segment analysis reveals model weaknesses
- Richer context for interpreting average metrics
- Diagnoses underlying model problems

EXAM TRAP: Error distributions reveal **patterns and biases** that averages hide!

T21: Model Comparison Example

TOPIC: Comparing Alternative Models

EXAMPLE: House Price Prediction

Model	MSE	MAE	MAPE
Model A (Linear Regression)	94.85	6.56	18.25
Model B (Decision Tree)	77.35*	5.41*	14.45*

CONCLUSION:

- Model B has all lower values
- Decision tree is more accurate according to all metrics
- Asterisk denotes best performer

KEY INSIGHT:

- Different metrics can yield different model rankings
- Choice depends on problem, regulatory situation, and historical precedent
- Lower values = better performance for all these metrics

EXAM TRAP: Lower values = **better performance** for all these metrics!

TOPIC: Why Different Metrics Can Yield Different Rankings

WHY RANKINGS CAN DIFFER:

- **MSE/RMSE:** Penalize large errors heavily
- **MAE:** Treats all errors equally
- **MAPE:** Emphasizes relative errors
- Different metrics = different priorities

IMPLICATION:

- Choose metric based on business needs
- Regulatory requirements may dictate choice
- Historical organizational choices matter
- No single "best" metric for all situations

EXAMPLE SCENARIO:

- Model A: Lower MAE but higher RMSE
- Model B: Lower RMSE but higher MAE
- Choice depends on whether tail risk or average error matters more

EXAM TRAP: Different metrics = **different rankings** - choose based on problem!

T23: Unnormalized Metrics Limitation

TOPIC: Limitations of Unnormalized Metrics

THE PROBLEM:

- RMSE and MAE are both unnormalized measures
- They treat errors symmetrically regardless of scale
- Example: \$2,000 error on \$150,000 salary (1.3%) vs \$2,000 error on \$12,000 salary (16.7%)
- Both contribute same to RMSE/MAE
- But the second error is much more significant proportionally

WHY THIS MATTERS:

- In some situations, a \$2,000 error on \$150,000 might be considered much less important
- Represents about 1% of total salary
- The other situation represents more than 10% of salary
- Unnormalized metrics don't capture this distinction

SOLUTION:

- Use MAPE for relative comparison
- Understand the context of errors
- Consider business impact, not just statistical measures

EXAM TRAP: Unnormalized metrics treat **all errors equally** - MAPE handles this issue!

TOPIC: ECB Recommendations on Performance Metrics

ECB GUIDE TO INTERNAL MODELS:

- Recommends use of metrics that capture tail risk
- Nudges firms to report RMSE alongside MAE
- Ensures that impact of large, infrequent losses is not overlooked

WHY THIS MATTERS:

- MAE alone might mask critical risks
- RMSE captures the impact of extreme events
- Both together provide complete picture
- Aligns with regulatory capital frameworks

REGULATORY CONTEXT:

- ECB's Guide to Internal Models
- Focus on tail risk and extreme events
- Capital adequacy requirements
- Systemic risk concerns

EXAM TRAP: ECB recommends **RMSE alongside MAE** for complete risk picture!

TOPIC: Why Auditors Prefer MAPE

AUDITOR PERSPECTIVE:

- Auditors often gravitate to MAPE
- Pure percentage narrative when discussing model accuracy with audit committees
- Provides standardized measure
- Allows comparison across different models/portfolios

WHY PERCENTAGE NARRATIVE MATTERS:

- Easy to understand for non-technical stakeholders
- Standardized across different scales
- Facilitates comparison
- Communicates effectively with audit committees

DIFFERENT STAKEHOLDER PREFERENCES:

- **Regulators:** RMSE alongside MAE (tail risk)
- **Auditors:** MAPE (percentage narrative)
- **Risk Managers:** All metrics for complete picture
- **Board:** MAPE and accuracy (intuitive)

EXAM TRAP: Auditors prefer MAPE for its **percentage narrative!**

TOPIC: The Confusion Matrix - Foundation of Classification Metrics

DEFINITION:

- Cross-tabulation of predicted classifications against actual outcomes
- Simple yet powerful tool providing clear breakdown of model performance
- Cornerstone for evaluating classification models

STRUCTURE:

	Predicted Negative	Predicted Positive
Actual Negative	True Negative (TN)	False Positive (FP)
Actual Positive	False Negative (FN)	True Positive (TP)

DEFINITIONS FROM RISK MANAGER'S PERSPECTIVE:

- **True Negative (TN)**: Model correctly identified negative case
- **False Positive (FP)**: Model incorrectly classified negative as positive (Type I Error)
- **False Negative (FN)**: Model incorrectly classified positive as negative (Type II Error)
- **True Positive (TP)**: Model correctly identified positive case

EXAM TRAP: Confusion matrix = **predicted vs actual** cross-tabulation!

T27: True Positive (TP)

TOPIC: True Positive - Correctly Identified Positive Case

DEFINITION:

- The model correctly identified a positive case
- Desired outcome - what we want the model to do

EXAMPLES:

- Correctly flagged a fraudulent transaction
- Correctly predicted that a loan would default
- Correctly identified a dividend-paying firm
- Correctly detected a critical compliance breach

WHY IT MATTERS:

- More TP = better model performance
- True positives represent value creation
- Catching fraud, defaults, or breaches avoids losses

EXAMPLE (Dividend Prediction):

- $TP = 432$
- 432 firms correctly predicted to pay dividends
- These are the successful predictions

EXAM TRAP: True Positive = correctly identified positive!

T28: True Negative (TN)

TOPIC: True Negative - Correctly Identified Negative Case

DEFINITION:

- The model correctly identified a negative case
- Desired outcome - what we want the model to do

EXAMPLES:

- Correctly identified a legitimate transaction
- Correctly predicted that a loan would not default
- Correctly identified a non-dividend paying firm
- Correctly cleared a compliant process

WHY IT MATTERS:

- More TN = better model performance
- True negatives represent efficiency
- Avoiding unnecessary interventions or investigations

EXAMPLE (Dividend Prediction):

- $TN = 279$
- 279 firms correctly predicted to not pay dividends
- These are the successful negative predictions

EXAM TRAP: True Negative = correctly identified negative!

T29: False Positive (FP)

TOPIC: False Positive - Type I Error

DEFINITION:

- The model incorrectly classified a negative case as positive
- Also known as Type I Error
- "False alarm"

EXAMPLES:

- Flagged a legitimate transaction as fraudulent
- Predicted that a good loan would default
- Incorrectly identified a firm as paying dividends
- Wrongly flagged a compliant process as breach

COST PERSPECTIVE:

- Operational cost to investigate false alert
- Customer friction from declined transactions
- Lost revenue from declining good customers
- Resource waste from chasing red herrings

EXAMPLE (Dividend Prediction):

- $FP = 121$
- 121 firms incorrectly predicted to pay dividends
- False alarms in prediction

EXAM TRAP: False Positive = Type I Error - incorrectly saying "yes"!

T30: False Negative (FN)

TOPIC: False Negative - Type II Error

DEFINITION:

- The model incorrectly classified a positive case as negative
- Also known as Type II Error
- "Missed detection"

EXAMPLES:

- Missed a fraudulent transaction (labeled it legitimate)
- Failed to predict that a loan would default
- Incorrectly identified a firm as not paying dividends
- Missed a critical compliance breach

COST PERSPECTIVE:

- Often the most costly type of error
- Financial losses from undetected fraud
- Capital impact of missed defaults
- Regulatory sanctions for missed breaches

EXAMPLE (Dividend Prediction):

- $FN = 168$
- 168 firms incorrectly predicted to not pay dividends
- Missed dividend payers

EXAM TRAP: False Negative = Type II Error - incorrectly saying "no"!

T31: Type I vs Type II Errors

TOPIC: Comparing Type I and Type II Errors in Credit Risk

WHY TYPE II ERRORS ARE MORE COSTLY IN CREDIT RISK:

- Misclassifying a borrower as solvent who then defaults
- Results in actual loan losses for the bank
- Regulatory capital implications
- Direct financial impact

TYPE I ERROR COST:

- Lost profit from not making a loan
- Operational cost of investigation
- Customer friction
- Usually less severe than Type II

GENERAL PRINCIPLE:

- The cost of errors is typically not symmetric
- Choose threshold based on relative costs
- In credit risk, Type II errors usually more costly
- In AML, both types have significant costs

EXAMPLES:

- **Credit Risk:** Type II more costly (missed default)
- **Fraud Detection:** Type II more costly (missed fraud)
- **AML:** Both costly (missed suspicious activity vs customer friction)

EXAM TRAP: Type II Errors (False Negatives) are **usually more costly** in credit risk!

T32: Accuracy Definition

TOPIC: Accuracy - Basic Classification Performance Measure

FORMULA:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

INTERPRETATION:

- "Percentage of all predictions that were correct"
- Sum of diagonal elements of confusion matrix divided by total
- Straightforward to interpret - simple agreement between predicted and observed

EXAMPLE (Dividend Prediction):

- Accuracy = $(432 + 279) / 1000 = 71.1\%$
- "71.1% of dividend predictions were correct"
- Error rate = $100\% - 71.1\% = 28.9\%$

LIMITATIONS:

- Doesn't distinguish between error types (FP and FN treated equally)
- Costs of errors are typically not symmetric
- Misleading for imbalanced datasets
- 98% positive, classify all as positive $\rightarrow$ 98% accuracy but useless model

EXAM TRAP: Accuracy = [percentage of all correct predictions!](#)

T33: Accuracy Example

TOPIC: Calculating Accuracy from Confusion Matrix

EXAMPLE DATA:

- True Positive (TP) = 432
- True Negative (TN) = 279
- False Positive (FP) = 121
- False Negative (FN) = 168

CALCULATION:

- $\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$
- $\text{Accuracy} = (432 + 279) / (432 + 279 + 121 + 168)$
- $\text{Accuracy} = 711 / 1000$
- $\text{Accuracy} = 71.1\%$

INTERPRETATION:

- "71.1% of all predictions were correct"
- Error rate = 28.9% (percentage of incorrect predictions)
- 711 out of 1000 predictions were correct

CONTEXT:

- 600 firms actually paid dividends
- 400 firms actually did not
- Model correctly classified 71.1% of all firms

EXAM TRAP: $\text{Accuracy} = (\text{TP} + \text{TN}) / \text{Total!}$

TOPIC: Error Rate and Its Relationship to Accuracy

FORMULA:

$$ErrorRate = 1 - Accuracy = \frac{FP + FN}{TP + TN + FP + FN}$$

INTERPRETATION:

- "Percentage of predictions that were incorrect"
- Complementary to accuracy
- Simple and intuitive

EXAMPLE:

- Accuracy = 71.1%
- Error Rate = 100% - 71.1% = 28.9%
- "28.9% of predictions were wrong"

RELATIONSHIP:

- Error Rate = 1 - Accuracy
- They are complementary measures
- Both ignore the type of error made
- Both problematic for imbalanced datasets

EXAM TRAP: Error rate = 1 - Accuracy!

TOPIC: Major Limitations of Accuracy

LIMITATION 1: DOESN'T DISTINGUISH ERROR TYPES

- FP and FN treated equally
- But costs are usually different
- Type II errors often more costly in credit risk

LIMITATION 2: MISLEADING FOR IMBALANCED DATA

- 98% positive, classify all as positive $\rightarrow$ 98% accuracy
- But model is practically useless
- Example: Fraud detection - 99% legitimate
- Model predicting "legitimate" always has 99% accuracy
- But catches zero fraud (Recall = 0)

LIMITATION 3: IGNORES COSTS

- Doesn't account for different costs of FP vs FN
- In credit risk, FN (missed default) much more costly

WHAT TO USE INSTEAD:

- Precision, Recall, F1 Score
- AUC (Area Under ROC Curve)
- Cost-sensitive metrics

EXAM TRAP: Accuracy is misleading for imbalanced datasets!

T36: Precision Definition

TOPIC: Precision - Of Predicted Positives, How Many Were Correct?

FORMULA:

$$Precision = \frac{TP}{TP + FP}$$

MANAGERIAL QUESTION:

- "Of all the alerts our model generated, how many were actually real issues?"
- "If we act on this prediction, how often will we be right?"

INTERPRETATION:

- Estimate of probability that model is right when labelling an outcome as true
- Of all instances classified as positive, what fraction were actually positive?

WHEN PRECISION MATTERS:

- High cost of investigation (AML alerts)
- Customer friction concerns
- Operational resource constraints
- When false alarms are expensive

EXAMPLE (Dividend Prediction):

- Precision = $432 / (432 + 121) = 432/553 = 78.1\%$
- "When model predicts dividend, it's right 78.1% of the time"

EXAM TRAP: Precision = $TP / (TP + FP)$ - "of predicted positives, how many were right?"

T37: Precision Example

TOPIC: Calculating Precision from Confusion Matrix

EXAMPLE DATA:

- True Positive (TP) = 432
- False Positive (FP) = 121

CALCULATION:

- Precision = $TP / (TP + FP)$
- Precision = $432 / (432 + 121)$
- Precision = $432 / 553$
- Precision = 78.1%

INTERPRETATION:

- "When the model predicts a firm will pay a dividend, it's right 78.1% of the time"
- "Of all dividend predictions, 78.1% were correct"

CONTEXT:

- 432 firms correctly predicted to pay dividends
- 121 firms incorrectly predicted to pay dividends
- Total predicted positive = 553
- Precision = 78.1%

EXAM TRAP: Precision = $TP / (TP + FP)$!

T38: Recall Definition

TOPIC: Recall (Sensitivity) - Of Actual Positives, How Many Did We Catch?

FORMULA:

$$\text{Recall} = \frac{TP}{TP + FN}$$

ALSO KNOWN AS: Sensitivity, True Positive Rate

MANAGERIAL QUESTION:

- "Of all real fraud events that occurred, how many did we catch?"
- "What percentage of actual positives did we identify?"

INTERPRETATION:

- Proportion of correctly classified positives among all positive instances
- True positive rate

WHEN RECALL MATTERS:

- Missing a positive has severe consequences
- Fraud detection, default prediction
- Compliance monitoring

SPECIFICITY (TRUE NEGATIVE RATE):

$$\text{Specificity} = \frac{TN}{TN + FP}$$

EXAMPLE (Dividend Prediction):

- $\text{Recall} = 432 / (432 + 168) = 432/600 = 72.0\%$
- "Of all firms that paid dividends, model caught 72%"

EXAM TRAP: Recall = $TP / (TP + FN)$ - "of actual positives, how many did we catch?"

TOPIC: Calculating Recall from Confusion Matrix

EXAMPLE DATA:

- True Positive (TP) = 432
- False Negative (FN) = 168

CALCULATION:

- $\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$
- $\text{Recall} = 432 / (432 + 168)$
- $\text{Recall} = 432 / 600$
- $\text{Recall} = 72.0\%$

INTERPRETATION:

- "Of all firms that actually paid dividends, the model caught 72%"
- "The model missed 28% of dividend payers"

CONTEXT:

- 600 firms actually paid dividends
- 432 were correctly predicted
- 168 were missed (False Negatives)
- $\text{Recall} = 72.0\%$

EXAM TRAP: Recall = $\text{TP} / (\text{TP} + \text{FN})$!

T40: Precision vs Recall Tradeoff

TOPIC: The Tradeoff Between Precision and Recall

THE TRADEOFF:

- **High threshold:** Higher precision, lower recall
- **Low threshold:** Lower precision, higher recall
- Cannot maximize both simultaneously

EXAMPLE:

- Threshold = 0.9: Only very confident predictions
- Precision: High (few false positives)
- Recall: Low (miss many actual positives)

BUSINESS CONTEXT DETERMINES BALANCE:

- **Fraud detection:** Prioritize recall (catch all fraud)
- **AML monitoring:** High recall critical
- **Credit approval:** Balance needed
- **Marketing:** Precision more important

F1 SCORE: Captures both in a single metric

EXAM TRAP: Precision and Recall have **tradeoff** - cannot maximize both simultaneously!

T41: F1 Score Definition

TOPIC: F1 Score - Balancing Precision and Recall

FORMULA:

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

WHY HARMONIC MEAN:

- Harmonic mean penalizes imbalance
- Low precision OR low recall → Low F1
- Better than arithmetic mean for this purpose
- If either precision or recall is low, F1 is low

INTERPRETATION:

- Bounded between 0 and 1
- Close to 1 = high precision and high recall
- Low score = poor precision, poor recall, or both

EXAMPLE (Dividend Prediction):

- Precision = 78.1%
- Recall = 72.0%
- $F1 = 2 \times (0.781 \times 0.720) / (0.781 + 0.720)$
- $F1 = 2 \times 0.5623 / 1.501 = 1.1246 / 1.501 = 74.9\%$

EXAM TRAP: F1 = harmonic mean of precision and recall!

T42: F1 Score Example

TOPIC: Calculating F1 Score from Precision and Recall

EXAMPLE DATA:

- Precision = 78.1% = 0.781
- Recall = 72.0% = 0.720

CALCULATION:

- $F1 = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$
- $F1 = 2 \times (0.781 \times 0.720) / (0.781 + 0.720)$
- $F1 = 2 \times 0.5623 / 1.501$
- $F1 = 1.1246 / 1.501$
- $F1 = 0.7494 = 74.9\%$

INTERPRETATION:

- F1 = 74.9%
- Balances precision and recall
- Good overall performance

EXAM TRAP: $F1 = 2 \times (P \times R) / (P + R)$!

T43: Specificity Definition

TOPIC: Specificity - True Negative Rate

FORMULA:

$$\text{Specificity} = \frac{TN}{TN + FP}$$

INTERPRETATION:

- "Of all actual negatives, what percentage did we correctly identify?"
- True Negative Rate
- Proportion of all negative outcomes correctly predicted as negative

RELATION TO RECALL:

- Recall = TPR (True Positive Rate)
- Specificity = TNR (True Negative Rate)
- Both are important for ROC curve
- $FPR = 1 - \text{Specificity}$

WHEN SPECIFICITY MATTERS:

- When false positives are costly
- Customer friction concerns
- Resource constraints for investigation

EXAM TRAP: Specificity = $TN / (TN + FP)$ - True Negative Rate!

T44: Imbalanced Data Problem

TOPIC: Why Accuracy is Misleading for Imbalanced Datasets

THE PROBLEM:

- Dataset: 98% solvent, 2% default
- Model predicts "solvent" for everyone
- Accuracy = 98% (seems great!)
- But model catches zero defaults

WHAT METRICS TO USE INSTEAD:

- **Precision:** 0% (no defaults predicted)
- **Recall:** 0% (caught zero actual defaults)
- **F1:** 0% (poor precision and recall)
- **AUC:** 0.5 (no discriminatory ability)

EXAMPLE:

- Fraud detection: 99% legitimate
- Model predicting "legitimate" always has 99% accuracy
- But catches zero fraud (Recall = 0)

LESSON:

- Accuracy is insufficient for imbalanced data
- Use Precision, Recall, F1, AUC
- Consider cost-sensitive metrics

EXAM TRAP: Accuracy is **misleading for imbalanced data!**

T45: Confusion Matrix Example

TOPIC: Complete Confusion Matrix Analysis

EXAMPLE DATA:

- True Positive (TP) = 60
- True Negative (TN) = 498
- False Positive (FP) = 98
- False Negative (FN) = 5

CALCULATIONS:

- Total = $60 + 498 + 98 + 5 = 661$
- Accuracy = $(60 + 498) / 661 = 558/661 = 84.4\%$
- Precision = $60 / (60 + 98) = 60/158 = 38.0\%$
- Recall = $60 / (60 + 5) = 60/65 = 92.3\%$
- Error Rate = $1 - 0.844 = 15.6\%$

INTERPRETATION:

- Model catches 92.3% of defaults (great!)
- But only 38% of default predictions are correct
- Many false alarms (98 false positives)

EXAM TRAP: Work through each formula carefully - don't mix up denominators!

T46: ROC Curve Definition

TOPIC: Receiver Operating Characteristic (ROC) Curve

DEFINITION:

- Graphical illustration of tradeoff between true and false positive rates against different choices of threshold Z
- Popular way to visualize classification model performance across all thresholds

ROC CURVE COMPONENTS:

- **Y-axis:** True Positive Rate (Recall)
- **X-axis:** False Positive Rate ($1 - \text{Specificity}$)
- **Points:** From varying decision threshold

THRESHOLD EXTREMES:

- **Threshold = 0:** All predicted positive
 - $\text{TPR} = 1$ (all positives caught)
 - $\text{FPR} = 1$ (all negatives misclassified)
 - Point: $(1, 1)$
- **Threshold = 1:** All predicted negative
 - $\text{TPR} = 0$ (no positives caught)
 - $\text{FPR} = 0$ (all negatives correct)
 - Point: $(0, 0)$

KEY INSIGHT:

- For values between 0 and 1, TPR increases when FPR increases
- Shows tradeoff between catching positives and avoiding false alarms

EXAM TRAP: ROC shows **TPR vs FPR** tradeoff across thresholds!

TOPIC: Area Under ROC Curve (AUC) - Interpretation

DEFINITION:

- Area under the ROC curve
- Measures overall ability of model to distinguish between positive and negative classes
- Threshold-independent measure

AUC VALUES AND INTERPRETATION:

- **AUC = 1.0:** Perfect discrimination
- **AUC = 0.5:** No discrimination (random guessing)
- **AUC < 0.5:** Worse than random (should invert predictions)
- **Higher AUC = better model**

PROBABILISTIC INTERPRETATION:

- Probability that a random positive ranks higher than a random negative
- Measures ranking ability of model
- Threshold-independent

EXAMPLE:

- AUC = 0.8 means 80% chance that a randomly chosen positive is ranked higher than a randomly chosen negative

EXAM TRAP: AUC = probability that positive ranks higher than negative!

TOPIC: What $AUC = 0.5$ Indicates

$AUC = 0.5$:

- No predictive ability
- Model cannot distinguish between classes
- Equivalent to random guessing
- Dashed diagonal line on ROC plot

WHAT IT MEANS:

- Model has no useful predictive information
- Positive and negative cases are equally likely to be ranked higher
- No better than flipping a coin

$AUC < 0.5$:

- Worse than random predictions
- Model performs worse than random draws
- Should invert predictions (swap positive/negative)

TYPICAL GOOD MODELS:

- AUC between 0.7 and 0.9
- $AUC > 0.9$: Excellent discrimination

EXAM TRAP: $AUC = 0.5$ = no predictive ability - random guessing!

TOPIC: What $AUC = 1.0$ Indicates

$AUC = 1.0$:

- Perfect predictive ability
- Perfect separation between classes
- Every positive has higher score than every negative
- There exists a threshold with perfect classification

CAUTION:

- $AUC = 1.0$ on test data is suspicious
- Likely indicates overfitting or data leakage
- Real-world data rarely has $AUC = 1.0$
- Should investigate immediately

WHAT TO CHECK:

- Is there data leakage from test to training?
- Is the model too complex?
- Are there duplicate records?
- Is there temporal leakage?

EXAM TRAP: $AUC = 1.0$ = perfect separation - be suspicious!

TOPIC: Interpreting ROC Curve Shape

BETTER ROC CURVE:

- Curve closer to upper-left corner
- Higher TPR, lower FPR
- More area under curve (higher AUC)

ROC CURVE INTERPRETATION:

- **Upper-left corner:** $TPR=1, FPR=0$ (perfect)
- **Diagonal line:** $TPR=FPR$ (random)
- **Lower-right corner:** $TPR=0, FPR=1$ (worst)

COMPARISON:

- Curve A with more area = better
- Curve that stays above and left = better
- AUC captures this visually

SAMPLE ROC CURVE:

- Model A: Curve closer to upper-left (better)
- Model B: Curve closer to diagonal (worse)
- Model A has higher AUC

EXAM TRAP: Better ROC curve = closer to upper-left corner!

TOPIC: How Threshold Changes Affect ROC Points

THRESHOLD EXTREMES:

- **Threshold = 0:** All predicted positive
 - $TPR = 1$ (all positives caught)
 - $FPR = 1$ (all negatives misclassified)
 - Point: (1, 1)
- **Threshold = 1:** All predicted negative
 - $TPR = 0$ (no positives caught)
 - $FPR = 0$ (all negatives correct)
 - Point: (0, 0)

AS THRESHOLD INCREASES:

- Points move from (1,1) toward (0,0)
- TPR decreases (catch fewer positives)
- FPR decreases (fewer false alarms)

AS THRESHOLD DECREASES:

- Points move from (0,0) toward (1,1)
- TPR increases (catch more positives)
- FPR increases (more false alarms)

EXAM TRAP: ROC goes from (0,0) to (1,1) as threshold decreases!

T52: AUC for Model Comparison

TOPIC: Using AUC to Compare Models in Lending Decisions

AUC IN LENDING:

- Measures ability to rank borrowers by risk
- Higher AUC = better separation of good from bad loans
- Better model = more accurate risk ranking

DECISION CONTEXT:

- Choose model with higher AUC on test set
- AUC = 0.8 vs 0.7: First has better ranking
- Better AUC = more confidence in decisions

HOW IT WORKS:

- Model ranks all loan applicants by default probability
- Higher AUC means better able to separate good from bad
- Can set threshold based on risk appetite

EXAMPLE:

- Model A: AUC = 0.85
- Model B: AUC = 0.72
- Model A better at ranking borrowers by risk
- Choose Model A for lending decisions

EXAM TRAP: Higher AUC = **better model** for lending decisions!

TOPIC: When to Use Precision-Recall Curve Instead of ROC

PRECISION-RECALL CURVES:

- Better for imbalanced datasets
- Focus on positive class performance
- More sensitive to rare events

WHY NOT ROC:

- ROC can be optimistic with imbalance
- Large number of negatives makes FPR low
- AUC can be high even with poor positive performance

WHEN TO USE EACH:

- **ROC:** Balanced datasets or when both classes equally important
- **Precision-Recall:** Imbalanced datasets, focus on positive class

EXAMPLE:

- Fraud detection (1% fraud): Use Precision-Recall
- Credit approval (50/50): Use ROC

EXAM TRAP: Precision-Recall for **imbalanced data!**

TOPIC: Extending Classification Metrics to Multi-class Problems

ONE-VS-REST APPROACH:

- For each class, treat as "positive"
- All other classes as "negative"
- Compute TP, FP, FN, TN for each class
- Average or aggregate metrics

EXAMPLE - CREDIT RATINGS:

- Classes: AAA, AA, A, BBB, BB, B, CCC
- For AAA: positive = AAA, negative = all others
- Compute precision/recall for AAA
- Repeat for each rating

GENERALIZATION:

- Precision: $\text{Sum all TP} / \text{Sum all TP} + \text{Sum all FP}$
- Recall: $\text{Sum all TP} / \text{Sum all TP} + \text{Sum all FN}$
- Other metrics generalized similarly

APPLICATION:

- Multi-class credit rating prediction
- Risk categorization (high/medium/low)
- Severity classification

EXAM TRAP: Multi-class uses **one-vs-rest** approach!

TOPIC: Advantages of AUC Over Accuracy

AUC ADVANTAGES:

- **Threshold-independent:** Evaluates all thresholds
- **Works with imbalance:** Not skewed by majority class
- **Ranking ability:** Measures how well model orders cases
- **Comprehensive:** Single number captures overall performance

ACCURACY LIMITATIONS:

- Threshold-dependent (uses one threshold)
- Misleading for imbalanced data
- Doesn't capture ranking ability
- Treats all errors equally

EXAMPLE:

- Dataset: 98% solvent, 2% default
- Model predicting all solvent: 98% accuracy
- But $AUC = 0.5$ (no discrimination)
- Accuracy misleading, AUC reveals truth

EXAM TRAP: AUC is **threshold-independent** - a key advantage!

TOPIC: Decision Threshold in Classification

DEFINITION:

- Most classification models output a probability or score
- A decision threshold is applied to convert this score into a binary classification
- Example: If PD $\geq$ 5%, classify as "high risk"; if fraud score $\geq$ 0.7, block transaction

COMMON THRESHOLDS:

- **Default:** 0.5 (balanced)
- **Can be adjusted:** Based on business needs
- **Different thresholds:** Different tradeoffs

WHY THRESHOLD CHOICE MATTERS:

- Directly determines balance of False Positives and False Negatives
- For every threshold, different Confusion Matrix
- Different Precision and Recall values
- Critical business decision, not just technical

EXAMPLE:

- Model outputs probability of default
- Threshold = 0.5: Predict default if PD $\geq$ 0.5
- Threshold = 0.3: More conservative, predict more defaults

EXAM TRAP: Threshold converts **probability to class label!**

TOPIC: Cost Matrix for Threshold Selection

DEFINITION:

- Explicitly defining economic consequences associated with each quadrant of Confusion Matrix
- Most effective way to guide threshold selection

COST MATRIX COMPONENTS:

- **Cost of False Negative:** Average loss from undetected fraud; capital impact of missed default (often largest cost)
- **Cost of False Positive:** Operational cost to investigate false alert; potential lost revenue from declining good customer
- **Benefit of True Positive:** Loss avoided by catching fraud; benefit of early intervention
- **Benefit of True Negative:** Smooth processing; benefit of correctly underwriting good loan

OPTIMAL THRESHOLD:

- Threshold that minimizes total expected cost
- OR maximizes total expected benefit
- Once costs/benefits estimated (even roughly)

EXAM TRAP: Cost matrix = **economic consequences** of each decision!

TOPIC: Cross-Functional Input for Cost Matrix

KEY STAKEHOLDERS:

- **Business Lines:** Understand cost of missed opportunities, customer attrition
- **Operations:** Know tangible costs of manual reviews (investigator time, call center)
- **Finance/Risk:** Can quantify financial impact of missed risks
- **Compliance/Legal:** Advise on potential costs of regulatory breaches

WHY MULTIPLE PERSPECTIVES:

- Different stakeholders understand different costs
- Complete picture of all consequences
- Collaborative decisions are more defensible
- Developing cost matrix is not purely quantitative exercise

EXAMPLES:

- Business: Cost of declining good loan
- Operations: Cost of AML alert investigation
- Finance: Capital impact of missed default
- Legal: Cost of regulatory breach

EXAM TRAP: Cost matrix needs **cross-functional input**!

T59: Dynamic Threshold Adjustment

TOPIC: Thresholds Should Be Dynamic, Not Static

DYNAMIC ADJUSTMENT EXAMPLES:

- **Holiday Fraud Spike:** Lower threshold (higher recall, catch more fraud)
- **Economic Downturn:** Adjust default thresholds
- **Benign Periods:** Higher threshold (fewer false positives)

PREREQUISITES:

- Clear risk appetite statement
- Documented review process
- Transparent justification
- Regulatory awareness

DOCUMENTATION EXAMPLE:

- "Threshold set for AML monitoring to achieve 80% recall"
- "Acceptable false positive rate of X%"
- "Deemed acceptable given consequences of missing AML events"

CAUTION:

- Ensure threshold adjusted for right reasons
- Not "set and forget"
- Actively managed

EXAM TRAP: Thresholds should be **dynamic** - not "set and forget"!

T60: Threshold Documentation

TOPIC: Documentation and Defensibility of Threshold Choices

DOCUMENTATION SHOULD SHOW:

- How threshold aligns with risk appetite
- Tradeoff between recall and false positives
- Costs and benefits considered
- Periodic review process

WHY IT MATTERS:

- Regulatory requirement
- Audit trail
- Knowledge transfer
- Defensible decisions

EXAMPLE STATEMENT:

- "We are currently setting threshold for transaction monitoring system to achieve 80% recall for high-risk AML typologies"
- "This means we accept a False Positive Rate of X%, costing Y in operational overhead"
- "Deemed acceptable given severe consequences of missing AML events"

PERIODIC REVIEW:

- Demonstrates not "set and forget"
- Actively managed in response to changing conditions
- Documented review process

EXAM TRAP: Thresholds must be **transparent and defensible!**

TOPIC: Costs Associated with False Positives in AML Monitoring

FALSE POSITIVE COSTS:

1 Operational Cost:

- Investigator time wasted
- Compliance teams chasing red herrings
- Resource consumption

2 Customer Friction:

- Declined legitimate transactions
- Customer inconvenience
- Potential customer attrition

3 Reputational Cost:

- Customer dissatisfaction
- Negative word-of-mouth

4 Alert Fatigue:

- Teams desensitized to alerts
- May miss real issues

WHY THIS MATTERS:

- High FP rate = inefficient operations
- Can outweigh benefits of catching fraud
- Affects customer experience

EXAM TRAP: False Positives = operational and customer costs!

TOPIC: Costs Associated with False Negatives in Default Prediction

FALSE NEGATIVE COSTS:

① Financial Losses:

- Actual loan losses
- Direct financial impact
- Hard to recover

② Regulatory Capital Implications:

- Capital add-ons
- Increased capital requirements
- Regulatory scrutiny

③ Risk Management:

- Misclassification of risk
- Underpricing of risk

④ Reputational Impact:

- Regulatory sanctions
- Market confidence erosion

WHY OFTEN LARGEST COST:

- Direct financial impact
- Hard to recover from
- Regulatory consequences

EXAM TRAP: False Negatives = financial and regulatory costs - usually largest!

TOPIC: Determining the Optimal Threshold Using Cost Matrix

EXPECTED COST CALCULATION:

- For each threshold, compute expected cost
- $\text{Cost} = C(TP) \times P(TP) + C(FP) \times P(FP) + C(FN) \times P(FN) + C(TN) \times P(TN)$
- Choose threshold with minimum cost
- Or maximum expected benefit

COST-BASED THRESHOLD:

- Different from accuracy-based threshold
- Reflects business reality
- More defensible

PROCESS:

- 1 Define costs/benefits for each quadrant
- 2 For each possible threshold, compute expected cost
- 3 Select threshold minimizing cost
- 4 Document decision

EXAM TRAP: Optimal threshold = minimum expected cost!

TOPIC: Which Model for Fraud Detection?

FRAUD DETECTION PRIORITIES:

- Missing fraud (False Negative) is very costly
- Better to catch more fraud (high recall)
- Accept some false alarms (lower precision)
- Business needs drive the tradeoff

OTHER CONTEXTS:

- **AML Monitoring:** High recall critical
- **Credit Approval:** Balance needed
- **Marketing:** Precision more important

EXAMPLE:

- Model A: High precision (90%), low recall (50%)
- Model B: Lower precision (60%), high recall (90%)
- For fraud detection, Model B preferred
- Catches more fraud despite more false alarms

EXAM TRAP: Fraud detection prioritizes **recall over precision!**

TOPIC: How Often Should Thresholds Be Reviewed?

REVIEW TRIGGERS:

- Significant environmental changes
- Shifts in risk appetite
- New regulatory requirements
- Changes in fraud patterns
- Economic downturns

BEST PRACTICE:

- Documented review schedule
- Clear triggers for review
- Stakeholder involvement
- Regulatory notification when appropriate

FREQUENCY:

- **Regular:** Quarterly, semi-annually, annually
- **Event-driven:** As needed based on triggers
- **Not:** Never or "set and forget"

EXAM TRAP: Thresholds should be **periodically reviewed** - not static!

T66: Model Comparison Example

TOPIC: Comparing Logistic Regression and Neural Network

EXAMPLE: LendingClub Default Prediction

Measure	LR Train	NN Train	LR Test	NN Test
Accuracy	0.842	0.842	0.713	0.701
Precision	0.656	0.750	0.600	0.541
Recall	0.236	0.169	0.283	0.377

KEY OBSERVATIONS:

- Performance drops from training to test (some overfitting)
- Different metrics give different rankings
- **Logistic Regression:** Better accuracy and precision on test
- **Neural Network:** Better recall on test
- No clear winner; choice depends on priorities

EXAM TRAP: Different metrics = different model rankings!

TOPIC: What Does Drop from Training to Test Indicate?

PERFORMANCE DROPS:

- Training accuracy: 84.2% → Test: 71.3% (LR)
- Training accuracy: 84.2% → Test: 70.1% (NN)
- Both show significant decrease

INTERPRETATION:

- Models overfit to training data
- Reasonable amount of overfitting
- Could improve with regularization
- Remove irrelevant features

REMEDICATION:

- "Remove some of the least empirically relevant features"
- "Apply regularization to the fitted models"

EXAM TRAP: Performance drop = **overfitting** - expected but should be managed!

TOPIC: Checking for Overfitting in Neural Networks

OVERFITTING CHECKS:

- 1 **Performance:** Training $\downarrow$ Test $\uparrow$
- 2 **Weights:** Very large or offsetting values
- 3 **Loss:** Training loss flat, validation increasing

WHAT TO LOOK FOR:

- Large positive and negative weights
- Model memorizing training data
- Poor generalization
- Strong or weak connections

EXAMPLE:

- Neural network contains ten units in hidden layer
- Uses logistic activation function
- Examine fitted weights for offsetting connections
- Indicative of overfitting

EXAM TRAP: Check **both performance gap and weight patterns** for overfitting!

TOPIC: Advantages of Logistic Regression Over Neural Networks

LOGISTIC REGRESSION ADVANTAGES:

- **Interpretability:** Coefficients explain feature impact
- **Explainability:** Easy to communicate to stakeholders
- **Speed:** Faster to train and deploy
- **Transparency:** Regulatory acceptance

NEURAL NETWORK ADVANTAGES:

- Can capture complex patterns
- Higher potential accuracy
- Better for large datasets

PRACTICAL CONSIDERATIONS:

- In LendingClub example, LR better on test accuracy and precision
- Simpler model may be preferred
- Regulatory requirements may favor LR

EXAM TRAP: Logistic regression = **more interpretable** than neural networks!

TOPIC: Advantages of Neural Networks Over Logistic Regression

NEURAL NETWORK ADVANTAGES:

- **Complex Patterns:** Can model nonlinear relationships
- **Flexibility:** Universal approximator
- **Performance:** Often better for complex tasks
- **Feature Learning:** Learns representations automatically

DISADVANTAGES:

- Less interpretable
- More data needed
- More computationally expensive
- Risk of overfitting

IN LENDINGCLUB EXAMPLE:

- Neural network had better recall (37.7% vs 28.3%)
- Caught more defaults
- But lower accuracy and precision
- Tradeoff between metrics

EXAM TRAP: Neural networks = **better for complex patterns** but less interpretable!

TOPIC: In LendingClub, Which Model Would a Risk Manager Choose?

TRADEOFFS:

- If catching defaults is critical: Neural network (higher recall)
- If interpretability is required: Logistic regression
- If accuracy is priority: Logistic regression (slightly higher)
- If regulatory compliance: Logistic regression

KEY INSIGHT:

- No single "best" model
- Choice depends on business needs
- Must align with risk appetite
- Contradictory indicators illustrate importance of choosing appropriate metric

DECISION FRAMEWORK:

- What is the cost of missing a default?
- What is the cost of false alarm?
- What are regulatory requirements?
- What is stakeholder preference?

EXAM TRAP: Model choice = depends on priorities and constraints!

TOPIC: How to Address Overfitting in LendingClub Example

REMEDIATION STRATEGIES:

- 1 **Simplify Model:** Remove irrelevant features
- 2 **Add Regularization:** L1, L2, dropout
- 3 **Get More Data:** More training samples
- 4 **Cross-Validation:** Better model selection
- 5 **Early Stopping:** Stop before overfitting

CHAPTER RECOMMENDATION:

- "Remove some of the least empirically relevant features"
- "Apply regularization to the fitted models"

WHY THIS WORKS:

- Reduces model complexity
- Prevents memorization
- Improves generalization

EXAM TRAP: Fix overfitting with **simplification or regularization!**

TOPIC: Which Metrics Should Be Used to Compare Models for Default Prediction?

RECOMMENDED APPROACH:

- 1 Start with confusion matrix
- 2 Compute Accuracy, Precision, Recall, F1
- 3 Consider business costs
- 4 Choose metric that aligns with priorities

WHY MULTIPLE METRICS:

- Single metric masks important differences
- Different stakeholders need different views
- Regulatory requirements may specify
- No single metric tells whole story

EXAMPLE:

- LR: Better accuracy and precision
- NN: Better recall
- Both have merits
- Choice depends on what matters most

EXAM TRAP: Use **multiple metrics** - don't rely on just one!

TOPIC: Why Test Sample Comparison is More Important Than Training

TRAINING SAMPLE:

- Shows how well model memorizes
- Optimistically biased
- Not representative of new data
- Not a true test of performance

TEST SAMPLE:

- Shows how well model generalizes
- Unbiased performance estimate
- What matters in practice
- True test of performance

KEY INSIGHT:

- We are generally more interested in validation or test sample
- Models overfit to training sample
- Examining accuracy on seen data is not true test

EXAM TRAP: Test sample = **true generalization** - always prioritize!

TOPIC: Tradeoff Between Logistic Regression and Neural Networks

LOGISTIC REGRESSION:

- **Pros:** Interpretable, transparent, regulatory friendly
- **Cons:** Limited to linear relationships

NEURAL NETWORKS:

- **Pros:** Can capture complex patterns
- **Cons:** Black box, hard to explain

CHOICE DEPENDS ON:

- Regulatory requirements
- Business needs
- Stakeholder expectations
- Interpretability vs performance

S&P GLOBAL STUDY:

- Identified tradeoffs between interpretability and performance
- Emphasized choosing models aligned with business objectives

EXAM TRAP: Tradeoff = **interpretability vs performance!**

TOPIC: Key Takeaway from LendingClub Model Comparison

KEY INSIGHTS:

- Different metrics gave different rankings
- Logistic regression: Better accuracy and precision
- Neural network: Better recall
- No clear winner

IMPLICATION:

- Model choice must align with business objectives
- "Contradictory indicators illustrate importance of choosing appropriate evaluation metric"
- Regulatory context matters
- No universal best model

PRACTICAL ADVICE:

- Understand what matters most
- Choose metrics accordingly
- Document rationale

EXAM TRAP: No universal best model - [choice depends on criteria!](#)

TOPIC: Why Evaluation Metrics Should Align with Business Objectives

WHY ALIGNMENT MATTERS:

- Metrics drive model selection
- Model selection drives decisions
- Decisions have real-world consequences
- Wrong metric = wrong decisions

EXAMPLES:

- If catching fraud is priority → optimize recall
- If minimizing investigations → optimize precision
- Must match business needs

ALIGNMENT PROCESS:

- 1 Identify business objectives
- 2 Determine cost of errors
- 3 Choose appropriate metrics
- 4 Align threshold with risk appetite

EXAM TRAP: Metrics must align with business objectives!

TOPIC: How Regulatory Requirements Affect Model Evaluation

REGULATORY EXPECTATIONS:

- ECB recommends tail-risk metrics
- RMSE alongside MAE
- Documentation of metric choice
- Justification for model selection

WHY IT MATTERS:

- Compliance is mandatory
- Penalties for non-compliance
- Reputational impact
- Regulatory scrutiny

KEY REGULATORY REQUIREMENTS:

- Metrics capturing tail risk
- RMSE alongside MAE
- Documentation and transparency
- Periodic review

EXAM TRAP: Regulators **specify metric requirements!**

TOPIC: Framework for Comparing Multiple Models

COMPARISON FRAMEWORK:

1 Multiple Metrics:

- Accuracy, Precision, Recall, F1, AUC
- Continuous: MSE, RMSE, MAE, MAPE

2 Business Context:

- Costs, priorities, risk appetite

3 Regulatory Requirements:

- Required metrics, documentation

4 Interpretability:

- Stakeholder needs

5 Performance:

- Training vs test gap

EXAM TRAP: Comprehensive comparison needs multiple perspectives!

TOPIC: Recommended Framework for Model Performance Evaluation

RECOMMENDED FRAMEWORK:

- ① Continuous Metrics:
 - MSE, RMSE, MAE, MAPE
- ② Classification Metrics:
 - Confusion Matrix, Accuracy, Precision, Recall, F1, AUC
- ③ Business Context:
 - Cost matrix, risk appetite
- ④ Documentation:
 - Transparent and defensible

KEY PRINCIPLE:

- No single metric tells the whole story
- Choose based on problem and stakeholders
- Document and justify choices

EXAM TRAP: Comprehensive evaluation = multiple metrics + business context + documentation!

TOPIC: What Error Distributions Reveal About Model Performance

KEY INSIGHTS FROM DISTRIBUTIONS:

- **Bias:** Systematic over/under prediction
- **Segments:** Areas where model performs poorly
- **Shape:** Many small vs few large errors

VISUALIZATIONS TO USE:

- Histograms of residuals
- Box plots by segment
- Q-Q plots for normality

KEY QUESTIONS:

- "Are errors generally small with a few large ones or consistently moderate?"
- "Are errors randomly distributed or systematically biased?"
- "Are there segments where errors are consistently larger?"

EXAM TRAP: Error distributions reveal **patterns and biases** that averages hide!

TOPIC: What Systematic Bias in Model Errors Indicates

SYSTEMATIC BIAS EXAMPLES:

- **Over-prediction:** Consistently predicting higher values
- **Under-prediction:** Consistently predicting lower values
- **Segment Bias:** Errors in specific subgroups

WHY IT MATTERS:

- Can lead to systematic business impact
- Harder to correct than random errors
- May indicate model misspecification
- More problematic than random errors even if average is small

EXAMPLE:

- Model consistently over-predicts recovery rates
- Leads to underestimation of losses
- Capital shortfall

EXAM TRAP: Systematic bias = consistent error pattern!

TOPIC: Why Analyze Performance Across Different Segments

BENEFITS OF SEGMENT ANALYSIS:

- Identifies model weaknesses
- Reveals fairness issues
- Highlights data quality problems
- Guides model improvements

EXAMPLE SEGMENTS:

- Geographic regions
- Product types
- Customer segments
- Economic conditions

WHAT TO LOOK FOR:

- Segments with consistently larger errors
- Segments with systematic bias
- Segments with different performance patterns

EXAM TRAP: Segment analysis reveals **where** models fail!

TOPIC: Why Regulators Prefer Metrics Capturing Tail Risk

TAIL RISK IMPORTANCE:

- Extreme events cause largest losses
- Capital requirements driven by tail risk
- Regulatory focus on solvency
- Systemic risk concerns

WHY MSE/RMSE:

- Heavily penalize large errors
- Capture tail events better than MAE
- Align with regulatory capital frameworks

ECB RECOMMENDATIONS:

- Metrics capturing tail risk
- RMSE alongside MAE
- Ensures large, infrequent losses not overlooked

EXAM TRAP: Regulators want [tail-risk metrics](#) for capital adequacy!

TOPIC: What Should Be Included in a Cost Matrix

COST COMPONENTS:

- 1 **Financial:** Losses from missed risks, capital impact
- 2 **Operational:** Investigation costs, resource waste
- 3 **Reputational:** Customer trust, market confidence
- 4 **Regulatory:** Fines, sanctions, scrutiny

BENEFIT COMPONENTS:

- **True Positive:** Losses avoided, risks managed
- **True Negative:** Efficient processing, satisfied customers

KEY INSIGHT:

- Cost matrix should capture all consequences
- Financial and non-financial
- Tangible and intangible

EXAM TRAP: Cost matrix should capture **all consequences** - financial and non-financial!

TOPIC: How to Justify Threshold Choices to Regulators

JUSTIFICATION SHOULD INCLUDE:

- 1 Analysis of tradeoffs
- 2 Cost-benefit calculations
- 3 Alignment with risk appetite
- 4 Stakeholder consultation
- 5 Periodic review process

EXAMPLE STATEMENT:

- "Threshold chosen to achieve 80% recall given severe consequences of missing fraud events"
- "Accepts X% false positives costing Y in operational overhead"

WHY IT MATTERS:

- Regulatory compliance
- Audit trail
- Defensible decisions

EXAM TRAP: Thresholds must be [justified with documented analysis](#)!

TOPIC: How to Communicate Model Performance to Different Stakeholders

AUDIENCE CONSIDERATIONS:

- **Board/Management:** MAPE, Accuracy (percentage-based)
- **Regulators:** RMSE with MAE, tail-risk metrics
- **Model Developers:** All metrics, error distributions
- **Audit Committee:** MAPE for standardized comparison

COMMUNICATION BEST PRACTICES:

- Use visualizations (ROC curves, error distributions)
- Explain what metrics mean in business terms
- Highlight tradeoffs and limitations

EXAM TRAP: Different stakeholders need **different metrics and explanations!**

TOPIC: How Often Should Model Performance Be Monitored?

MONITORING FREQUENCY:

- **Regular Intervals:** Quarterly, semi-annually, annually
- **Event-Driven:** Significant environmental changes
- **Trigger-Based:** Performance threshold breaches

WHAT TO MONITOR:

- Performance metrics over time
- Error distributions
- Segment performance
- Threshold appropriateness

WHY CONTINUOUS MONITORING:

- Detect performance decay
- Identify concept drift
- Ensure ongoing compliance

EXAM TRAP: Model performance must be **monitored continuously** - not just during development!

TOPIC: Role of Validation in Model Performance Evaluation

VALIDATION PURPOSES:

- 1 Independent performance assessment
- 2 Hyperparameter tuning
- 3 Overfitting detection
- 4 Model comparison

VALIDATION BEST PRACTICES:

- Use separate validation data
- Use cross-validation
- Document validation process
- Regulatory compliance

KEY INSIGHT:

- Validation provides independent assessment
- Different from training
- Essential for model governance

EXAM TRAP: Validation provides **independent assessment** of model performance!

TOPIC: Why Performance Metric Choices Should Be Documented

DOCUMENTATION SHOULD INCLUDE:

- 1 Choice of metrics
- 2 Justification for choices
- 3 Business context
- 4 Regulatory requirements
- 5 Performance results
- 6 Model selection rationales

WHY IT MATTERS:

- Regulatory compliance
- Audit trail
- Knowledge transfer
- Defensible decisions

EXAM TRAP: Metric choices must be **documented and justified!**

TOPIC: How to Detect Model Performance Decay

DETECTION METHODS:

- Track metrics over time
- Compare to baseline performance
- Monitor error distributions
- Segment performance tracking

COMMON CAUSES:

- Data drift (concept drift)
- Economic changes
- Regulatory changes
- Model degradation

WHY IT MATTERS:

- Early detection allows remediation
- Prevents silent model failure
- Maintains performance standards

EXAM TRAP: Monitor metrics over time to detect **performance decay**!

TOPIC: What is Concept Drift and How Does It Affect Performance?

DEFINITION:

- Changes in underlying relationship between inputs and outputs over time
- Model trained on historical data may become less accurate

EXAMPLES:

- Changing fraud patterns
- Economic cycles affecting default relationships
- Customer behavior changes
- Regulatory changes

DETECTION AND RESPONSE:

- Monitor performance metrics
- Regularly retrain models
- Update model parameters
- Review thresholds

EXAM TRAP: Concept drift = [changing relationships](#) over time - requires monitoring!

TOPIC: What Should Trigger Model Retraining?

RETRAINING TRIGGERS:

- 1 **Performance:** Metrics below acceptable levels
- 2 **Data:** Significant changes in data distribution
- 3 **Schedule:** Regular intervals (monthly, quarterly)
- 4 **Events:** Major economic or regulatory changes

BEST PRACTICE:

- Document retraining triggers
- Regular performance review
- Proactive monitoring
- Automated alerts

KEY INSIGHT:

- Models are not permanent
- Need periodic refresh
- Align with business changes

EXAM TRAP: Retrain when **performance decays**, **data changes**, or **on schedule**!

TOPIC: What Should Be Included in Model Performance Reports

REPORT COMPONENTS:

- ❶ **Current:** Latest performance metrics
- ❷ **Trends:** Performance over time
- ❸ **Baseline:** Comparison to initial/expected performance
- ❹ **Issues:** Any identified problems or concerns
- ❺ **Actions:** Planned or taken remediation

AUDIENCE:

- Model owners
- Management
- Regulators
- Audit committees

KEY INSIGHT:

- Reports should include current metrics, trends, and issues
- Regular and transparent

EXAM TRAP: Reports should include **current metrics, trends, and issues!**

TOPIC: How Model Performance Evaluation Helps Identify Fairness Issues

FAIRNESS EVALUATION:

- Compare performance across segments
- Check for systematic bias
- Ensure equal treatment
- Regulatory compliance (fair lending)

WHAT TO EXAMINE:

- Accuracy differences
- False positive/negative rates
- Disparate impact
- Equal opportunity

WHY IT MATTERS:

- Legal and regulatory requirements
- Ethical considerations
- Reputational risk

EXAM TRAP: Segment analysis reveals fairness issues!

TOPIC: How Model Performance Evaluation Relates to Explainable AI

CONNECTION TO XAI:

- Poor performance in segments → need explanations
- Complex models (DNNs) → need XAI techniques
- Regulatory requirements for explanation
- Tools: SHAP, LIME for interpretation

PRACTICAL APPLICATION:

- Use explanations to understand poor performance
- Identify feature importance
- Debug model issues
- Build trust in predictions

EXAM TRAP: Performance evaluation guides [where explanations are needed!](#)

TOPIC: How SHAP Values Help with Model Performance Evaluation

SHAP IN EVALUATION:

- Identify important features
- Understand model behavior
- Debug performance issues
- Explain to stakeholders

APPLICATION:

- Feature importance analysis
- Understanding model decisions
- Regulatory explanations
- Model improvement insights

KEY INSIGHT:

- SHAP helps understand what drives model performance
- Identifies which features are driving good and bad performance

EXAM TRAP: SHAP helps understand what drives model performance!

TOPIC: When LIME is Useful for Model Evaluation

LIME APPLICATIONS:

- Explain specific predictions
- Understand why model made certain decisions
- Identify problematic cases
- Build trust with stakeholders

EVALUATION INSIGHTS:

- Why certain cases were misclassified
- What features drove errors
- Edge case behavior
- Model limitations

KEY INSIGHT:

- LIME explains individual predictions
- Useful for error analysis
- Complements global metrics

EXAM TRAP: LIME explains **individual predictions** - useful for error analysis!

TOPIC: Role of Model Governance in Performance Evaluation

GOVERNANCE COMPONENTS:

- 1 **Standardization:** Consistent metrics across models
- 2 **Documentation:** Clear records of evaluation
- 3 **Monitoring:** Regular performance review
- 4 **Oversight:** Independent validation

WHY IT MATTERS:

- Regulatory compliance
- Risk management
- Consistent decision-making
- Auditability

KEY INSIGHT:

- Governance ensures consistent evaluation and monitoring
- Essential for model risk management

EXAM TRAP: Governance ensures consistent evaluation and monitoring!

TOPIC: Key Takeaways from Model Performance Evaluation

KEY TAKEAWAYS:

- 1 **Multiple Metrics:** Use continuous (MSE, RMSE, MAE, MAPE) and classification (Confusion Matrix, Accuracy, Precision, Recall, F1, AUC)
- 2 **Business Context:** Align with business objectives and risk appetite
- 3 **Continuous Monitoring:** Track performance over time
- 4 **Documentation:** Record and justify all choices

CRITICAL INSIGHTS:

- No single metric tells the whole story
- Different metrics serve different purposes
- Performance evaluation is a continuous process
- Regulatory requirements must be met

FINAL ADVICE:

- Understand metrics, tradeoffs, and when to use each
- Choose based on problem and stakeholders
- Document and justify choices

EXAM TRAP: Comprehensive evaluation = multiple metrics + business context + monitoring + documentation!

Summary: Complete Review

Continuous Output Metrics:

- **MSE:** Average squared error; penalizes large errors; captures tail risk
- **RMSE:** Square root of MSE; same units as target; easier to interpret
- **MAE:** Average absolute error; robust to outliers
- **MAPE:** Percentage error; board-friendly; compares across scales

Classification Metrics:

- **Confusion Matrix:** TP, TN, FP, FN - foundation of all metrics
- **Accuracy:** Total correct / total - misleading for imbalanced data
- **Precision:** $TP / (TP + FP)$ - "of predicted positives, how many were right?"
- **Recall:** $TP / (TP + FN)$ - "of actual positives, how many did we catch?"
- **F1:** Harmonic mean of precision and recall
- **AUC:** Area under ROC curve - threshold-independent ranking ability

Threshold Selection:

- **Cost Matrix:** Economic consequences of each decision
- **Tradeoff:** Balance false positives vs false negatives
- **Dynamic:** Adjust based on changing conditions
- **Documentation:** Transparent and defensible

Exam Success: Understand the metrics, their tradeoffs, and when to use each!